

The eight-study portfolio: the question each study asks

grouped by theme; the question is the study's own and the status is the state of the evidence in this thesis

Scaling <i>does the recipe follow a law?</i> P1 · P3	P1 Scaling laws for GRPO post-training <i>Does held-out GRPO gain follow a predictable cross-scale slope, or does the recipe resist a scaling law?</i> complete · null recovered: no reliable cross-scale slope, and no identifiable saturation exponent	P3 Group size, contrast, DPO bridge <i>How do collapse, reward, held-out gain and token cost move with group size G — and what does G buy?</i> complete · under saturation, training-reward ratio ≈ 1.00 across G (equivalence to $G=8$ only); SNR is 52% of the $\sqrt{G}$ ideal
Diagnostic <i>what is the signal doing?</i> P2	P2 The Zero-Variance Fraction as a descriptive diagnostic <i>What fraction of each rollout batch supplies no reward-relative gradient, and can a collapsed group be re-baselined?</i> complete · recomputable from stored reward tensors (0.130, 0.190, 0.155; pooled 0.158); link to outcome $\rho \approx 0.27$	
Mechanism <i>why does the signal vanish?</i> P4 · P7	P4 Length bias and held-out generalisation <i>Do length penalties change how long a model answers — and does that change held-out accuracy at all?</i> complete · held-out gains indistinguishable; pooled length 194.4 $\rightarrow$ 183.5 tokens under a 200-token cap	P7 From diagnostic to controller <i>Can the ZVF diagnostic be closed into a controller that adapts group size from measured starvation?</i> partial · model and 368-run audit complete; the controller is not joined to P5
Reporting <i>what must be stated?</i> P5 · P6	P5 Report the stack, not the label <i>What minimum set of fields must a stack-conditioned result report before it can be attributed at all?</i> complete · flagship: the seven-field minimum-report manifest; badge 43–96 over twenty seed entries	P6 GRPO-Registry: a stack catalogue <i>Can evolving run state be recorded as a machine-readable registry, and diffed to predict a label flip?</i> complete · <i>stackdiff</i> flags the open-vs-closed DAPO pair as an R5 label-flip risk from manifests alone
Applied <i>does it transfer?</i> P8	P8 LLM vs. gradient-boosted trees in credit-card fraud <i>When two models are measured on different splits, what can the comparison actually support?</i> partial · archived side study absorbed as a measurement-discipline case study; the two AUCs are not like-for-like	

 complete — the study answers its question, including when the answer is a null

 partial — scope narrowed or a limb deliberately left unclaimed

Reading the map

Established

- The ZVF is measurable from stored reward tensors and reproduces: 0.130, 0.190, 0.155.
- Under saturation, the training-reward ratio is equivalent to $G=2$ up to $G=8$; equivalence is not established at $G=16$ ($p=0.056$).
- A seven-field manifest suffices to flag label-flip risk.

Declared null

- P1 recovers no reliable cross-scale slope and no identifiable saturation exponent.
- P4 finds GRPO and Dr. GRPO held-out gains indistinguishable.
- Its error-bar audit, re-graded, returns no decisive verdict and one suggestive.

Not claimed

- P7's controller is audited but not joined to P5's attribution programme.
- P8's two AUCs are not a like-for-like ranking: the splits differ by construction.
- P2's diagnostic-to-outcome association is not distinguishable from zero ($\rho=0.27$, CI $[-0.37, 0.88]$).